

考研竞赛题集. 第十四周. 同考.



1. 设 A, B 均为 n 阶正交矩阵. $|A| + |B| = 0$. 则 $|A+B| = \underline{\hspace{2cm}}$. (Ans: 0).

[解]: $\begin{cases} AA^T = A^T A = E \\ BB^T = B^T B = E \end{cases} \Rightarrow \begin{cases} |A|^2 = 1 \\ |B|^2 = 1 \end{cases}$. 对于 $|A| + |B| = 0$. 不妨设 $|A| = 1, |B| = -1$.

$$\begin{aligned} |A+B| &= -|A| |A+B| |B| = -|A^T| |A+B| |B^T| \\ &= -|A^T A + A^T B| \cdot |B^T| = -|\underline{A^T A} \cdot \underline{B^T} + \underline{A^T B} \cdot \underline{B^T}| \\ &= -|B^T + A^T| = -|(B+A)^T| = -|A+B| \end{aligned}$$

$$\Rightarrow |A+B| = 0$$





2. $\lim_{x \rightarrow 0} \int_0^{\frac{x}{2}} dv \int_v^{\frac{x}{2}} \frac{e^{(u-v)^2}}{1 - \cos x} du$ (Ans: $\frac{1}{4}$).

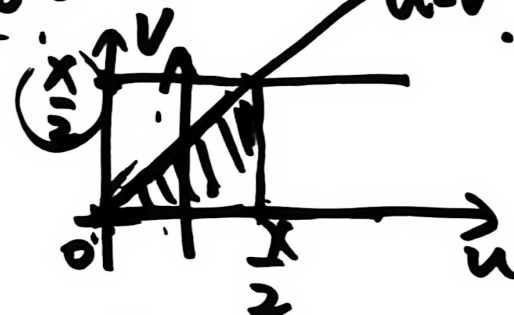
[1/27] $I = \lim_{x \rightarrow 0} \frac{\int_0^{\frac{x}{2}} dv \int_v^{\frac{x}{2}} e^{(u-v)^2} du}{1 - \cos x \sim \frac{1}{2} x^2}$

换元 $\lim_{x \rightarrow 0} \frac{\int_0^{\frac{x}{2}} du \int_0^u e^{(u-v)^2} dv}{\frac{1}{2} x^2}$

洛 $\lim_{x \rightarrow 0} \frac{\int_0^{\frac{x}{2}} e^{(\frac{x}{2}-v)^2} dv \cdot \frac{1}{2}}{\frac{1}{2} x^2}$

$= \lim_{x \rightarrow 0} \frac{\int_0^{\frac{x}{2}} e^{v^2} dv \cdot \frac{1}{2}}{x} \stackrel{\text{洛}}{=} \lim_{x \rightarrow 0} \frac{e^{\frac{x^2}{4}} \cdot \frac{1}{2}}{1} = \frac{1}{4}$

$(\int_0^x f(t) dt)' = f(x)$
 $u=v$



$(\int_0^{\frac{x}{2}} g(u) du)' = g(\frac{x}{2}) \cdot \frac{1}{2}$

$\frac{e^{\frac{x^2}{4}} \cdot \frac{1}{2}}{1} = \frac{1}{4}$

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3. $\int_0^{+\infty} \frac{\arctan x}{(1+x)^2} dx$ (Ans: $\frac{\pi}{4}$).



[Ans]: $I = - \int_0^{+\infty} \arctan x \cdot d \frac{1}{1+x} = - \frac{\arctan x}{1+x} \Big|_0^{+\infty} + \int_0^{+\infty} \frac{1}{1+x} \cdot \frac{1}{1+x^2} dx$

$= \int_0^{+\infty} \frac{1}{(1+x)(1+x^2)} dx \xrightarrow{x=\tan t} \int_0^{\frac{\pi}{2}} \frac{\cancel{\sec^2 t}}{(1+\tan t) \cdot \cancel{\sec^2 t}} dt$

$= \int_0^{\frac{\pi}{2}} \frac{\cos t}{\cos t + \sin t} dt = \int_0^{\frac{\pi}{2}} \frac{\sin t}{\sin t + \cos t} dt$

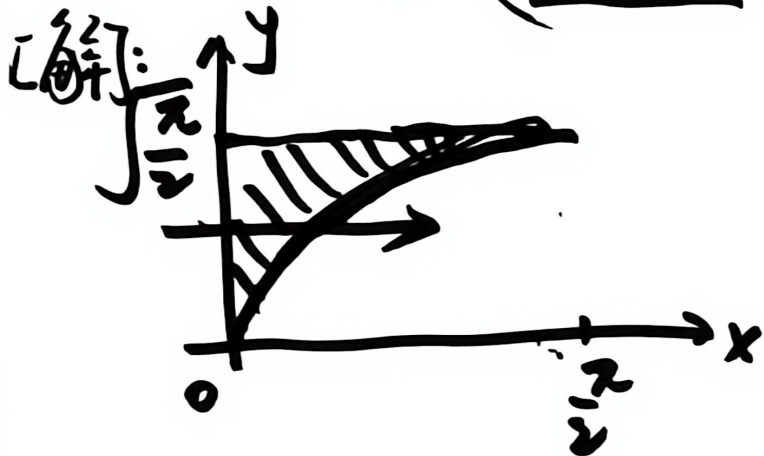
$= \frac{1}{2} \int_0^{\frac{\pi}{2}} 1 dt = \frac{1}{2} \cdot \frac{\pi}{2} = \frac{\pi}{4} \checkmark$



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4. $\int_0^{\frac{\sqrt{2}}{2}} \left(\frac{1}{\sqrt{x}} \right) dx \left(\frac{\sqrt{\frac{x}{2}}}{\sqrt{x}} \right) \left(\frac{1}{1 + (\tan y^2)^{\sqrt{2}}} \right) dy$ (Ans: $\frac{\pi}{4}$. 本段如上-是很像).
 $(\sqrt{x})' = \frac{1}{2\sqrt{x}}$



$x = y^2$

$$\Rightarrow I = \int_0^{\frac{\sqrt{2}}{2}} \frac{1}{\sqrt{x}} dx \int_0^{\frac{\sqrt{2}}{2}} \frac{1}{1 + (\tan y^2)^{\sqrt{2}}} dy$$

$$= 2 \cdot \int_0^{\frac{\sqrt{2}}{2}} \frac{1}{1 + (\tan y^2)^{\sqrt{2}}} \cdot [(y) - 0] dy$$

$$\frac{y^2}{2} \int_0^{\frac{\sqrt{2}}{2}} \frac{1}{1 + (\tan t)^{\sqrt{2}}} dt = \int_0^{\frac{\sqrt{2}}{2}} \frac{\cos^{\sqrt{2}} t}{\cos^{\sqrt{2}} t + \sin^{\sqrt{2}} t} dt$$

$$= \frac{1}{2} \cdot \int_0^{\frac{\sqrt{2}}{2}} 1 dt = \frac{\pi}{4} \checkmark$$



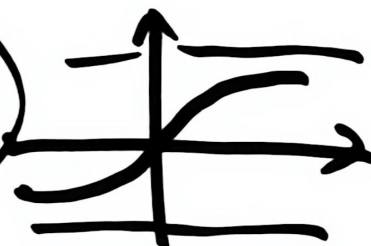
5. $\lim_{n \rightarrow \infty} \sum_{k=1}^n \frac{\sqrt{k^2+k}}{n^2 + \frac{1}{k}} \cdot \arctan^2\left(\frac{k}{n}\right)$. (Ans: $\frac{\pi^2}{16} - \frac{2}{4} + \frac{1}{2} \ln 2$).

[7/3]:

(1) $a_n \leq \sum_{k=1}^n \frac{k + \frac{1}{2}}{n^2} \cdot \arctan^2 \frac{k}{n}$

$$k^2 + k = \left(k + \frac{1}{2}\right)^2 - \frac{1}{4}.$$

$$= \frac{1}{n} \cdot \sum_{k=1}^n \frac{k}{n} \cdot \arctan^2 \frac{k}{n} + \frac{1}{2n^2} \cdot \sum_{k=1}^n \arctan^2 \frac{k}{n}$$



其中, $-\frac{\pi}{2} < \arctan x < \frac{\pi}{2} \Rightarrow \frac{1}{2n^2} \cdot \sum_{k=1}^n \arctan^2 \frac{k}{n} \leq \frac{2}{n^2} \cdot \sum_{k=1}^n \frac{\pi^2}{4}$

$$= \frac{\pi^2}{2n^2} \cdot n \xrightarrow{n \rightarrow \infty} 0$$

②.



例. $\lim_{n \rightarrow \infty} \frac{1}{n} \cdot \sum_{k=1}^n \frac{k}{n} \arctan^2 \frac{k}{n} = \int_0^1 x \arctan^2 x \, dx$

$$= \frac{1}{2} \int_0^1 \arctan^2 x \cdot d(x^2) = \frac{1}{2} \cdot \frac{x^2}{16} - \int_0^1 x^2 (\arctan x) \cdot \frac{1}{1+x^2} dx$$

$$= \frac{x^2}{32} - \int_0^1 \frac{x^2+1}{x^2+1} \cdot \arctan x \, dx$$

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$$= \frac{x^2}{32} - \int_0^1 \arctan x \, dx + \frac{1}{2} \cdot (\arctan x)^2 \Big|_0^1$$

$$= \frac{x^2}{32} - \frac{x}{4} + \int_0^1 x \cdot \frac{1}{1+x^2} dx + \frac{1}{2} \cdot \left[\frac{x^2}{16} - 0 \right] \ln(1+x^2) \Big|_0^1$$

$$= \frac{x^2}{16} - \frac{x}{4} + \frac{1}{2} \cdot \ln 2 \quad \checkmark$$



(2). $a_n \geq \sum_{k=1}^n \frac{k}{n^2+1} \arctan^2 \frac{k}{n}$

$\lim_{n \rightarrow \infty} \left(\frac{1}{n^2+1} \right) \sum_{k=1}^n k \cdot \arctan^2 \frac{k}{n}$

此例 $\lim_{n \rightarrow \infty} \left(\frac{1}{n^2} \right) \sum_{k=1}^n k \cdot \arctan^2 \frac{k}{n} = \int_0^1 x \arctan^2 x dx = \dots$

综上. 由夹逼. 原式 = $\frac{\pi^2}{16} - \frac{\pi^2}{4} + \frac{1}{2} \ln^2$

6. 设 $f(x,y) = F(\frac{y}{x})$. 且 $\frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial y^2} = 0$. 其中 $F(u) \neq 0$. 求 $f(x,y)$.



解: $u = \frac{y}{x}$ $F(u)$. $u = \frac{y}{x}$

$\frac{F'(u)}{x^2}$

$$\frac{\partial f}{\partial x} = F'(u) \cdot \left(-\frac{y}{x^2}\right) \Rightarrow \frac{\partial^2 f}{\partial x^2} = (-y) \cdot \frac{F''(u) \cdot \left(-\frac{y}{x^2}\right) \cdot x - F'(u) \cdot 2x}{x^4}$$

$$= y \frac{y \cdot F''(u) + 2x F'(u)}{x^4} = \frac{y^2 F''(u) + 2xy F'(u)}{x^4}$$

$$f(x,y) = C_1 \arctan \frac{y}{x} + C_2$$

$$\frac{\partial f}{\partial y} = F'(u) \cdot \frac{1}{x} \Rightarrow \frac{\partial^2 f}{\partial y^2} = \frac{1}{x} \cdot F''(u) \cdot \frac{1}{x} = \frac{F''(u)}{x^2}$$

$$\Rightarrow \cancel{\frac{1}{x^4}} [(x^2 y^2) \cdot F''(u) + 2xy F'(u)] = 0 \Rightarrow \left[1 + \left(\frac{y}{x}\right)^2 \right] \cdot F''(u) + 2 \frac{y}{x} \cdot F'(u) = 0$$

$$\Rightarrow (1+u^2) \cdot F''(u) + 2u F'(u) = 0 \Rightarrow [F'(u) \cdot (1+u^2)]' = 0 \Rightarrow F'(u) = \frac{C_1}{1+u^2} + C_2$$



7. 用 $x=e^t$ 代换 $(x^2 \ln x) \cdot y'' - xy' + y = 0$. 并求该微分方程通解

[解]: (目标: 求出 $\frac{dy}{dt}$, $\frac{d^2y}{dt^2}$, 及其微分方程).

$$x=e^t \Rightarrow t=\ln x$$

$$\frac{dy}{dx} = \frac{\frac{dy}{dt}}{\frac{dx}{dt}} = \frac{dy}{dt} \cdot \frac{1}{x} \Rightarrow \frac{dy}{dt} = x \cdot y'$$

$$\Rightarrow 1 \cdot y' + x \cdot y'' = \frac{d(\frac{dy}{dt})}{dx} = \frac{d(\frac{dy}{dt})}{dt} \cdot \frac{dt}{dx} = \frac{d^2y}{dt^2} \cdot \frac{1}{x} \Rightarrow \frac{d^2y}{dt^2} = xy' + x^2 y''$$

~~代入原方程~~ $t \cdot (\frac{d^2y}{dt^2} - \frac{dy}{dt}) - \frac{dy}{dt} + y = 0$

$$\Rightarrow t \cdot (\frac{d^2y}{dt^2} - \frac{dy}{dt}) - (\frac{dy}{dt} - y) = 0 \Rightarrow \left(\frac{\frac{dy}{dt} - y}{t} \right)' = 0$$

$$\Rightarrow \frac{\frac{dy}{dt} - y}{t} = C_1 \Rightarrow \frac{dy}{dt} - y = C_1 t \Rightarrow y = e^t \left[\int C_1 t \cdot e^{-t} dt + C_2 \right]$$

$$\text{则} \int t \cdot e^{-t} dt = - \int t d e^{-t} = -t e^{-t} + \int e^{-t} dt \\ = -t e^{-t} - e^{-t} + C.$$

$$\Rightarrow y = e^t \cdot [C_1 (-t e^{-t} - e^{-t}) + C_2]$$

$$= (-C_1) \cdot (t+1) + C_2 e^t$$

$$= -C_1 \cdot (\ln x + 1) + C_2 \cdot x$$

$$= C_3 \cdot (1 + \ln x) + C_2 x$$

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8. 设 $x > 0$ 且 $(1+ax)(\ln(1+x)) > x$ 恒成立, 求 a 的范围.



【解】: $\Leftrightarrow a > \left(\frac{x}{\ln(1+x)} - 1 \right) \cdot \frac{1}{x} = \frac{1}{\ln(1+x)} - \frac{1}{x} \Rightarrow a > \left[\frac{1}{\ln(1+x)} - \frac{1}{x} \right]_{x \rightarrow 0^+}$

令 $f(x) = \frac{1}{\ln(1+x)} - \frac{1}{x}$, $f'(x) = -\frac{1}{\ln^2(1+x)} \cdot \frac{1}{1+x} + \frac{1}{x^2} = \frac{(1+x)(\ln^2(1+x)) - x^2}{x^2 \cdot (1+x) \ln^2(1+x)} < 0$

令 $g(x) = (1+x)(\ln^2(1+x)) - x^2$, $g(0) = 0$

$g'(x) = 1 \cdot \ln^2(1+x) + 2 \ln(1+x) - 2x$, $g'(0) = 0$

$g''(x) = 2 \ln(1+x) \cdot \frac{1}{1+x} + 2 \cdot \frac{1}{1+x} - 2 = \frac{2}{1+x} \cdot [\ln(1+x) + 1 - x] < 0$

$g(x) = g(0) + g'(0)x + \frac{g''(\xi)}{2!} (x^2)^{>0} < 0 \Rightarrow f'(x) < 0 \Rightarrow f(x) \downarrow \Rightarrow f(x) < \lim_{x \rightarrow 0^+} f(x)$

$= \frac{1}{x \rightarrow 0} \left[\frac{1}{\ln(1+x)} - \frac{1}{x} \right] = \lim_{x \rightarrow 0} \frac{x - \ln(1+x)}{x \ln(1+x)} = \frac{1}{2} \Rightarrow a > \frac{1}{2} \checkmark$



9. 设 A 为 3 阶矩阵. $A^2 - A = 2E$. $\text{tr}(A) = 3$ 求 $|A^* + 2A^{-1}| = \underline{\hspace{2cm}}$.

[解] $A^2 - A - 2E = 0 \quad f(A) = 0 \Rightarrow f(\lambda) = 0$

$\Rightarrow \lambda^2 - \lambda - 2 = 0 \Rightarrow (\lambda - 2)(\lambda + 1) = 0 \Rightarrow \lambda = 2 \text{ 或 } -1$

$\text{tr}(A) = a_{11} + a_{22} + a_{33} = \lambda_1 + \lambda_2 + \lambda_3 = 3$

取 $A = \begin{pmatrix} 2 & & \\ & 2 & \\ & & -1 \end{pmatrix}$
 $A^* = \begin{pmatrix} 2 & & \\ & 2 & \\ & & 4 \end{pmatrix}$
 $2A^{-1} = \begin{pmatrix} 1 & & \\ & 1 & \\ & & -2 \end{pmatrix}$

另解: $|A| \cdot A^{-1} + 2A^{-1} = (-2) \cdot A^{-1}$
 $= (-2)^3 \cdot \frac{1}{|A|} = -8 \cdot \frac{1}{-4} = 2 \quad \checkmark$

$A^* + 2A^{-1}: -1, -1, 2 \Rightarrow |A^* + 2A^{-1}| = (-1) \cdot (-1) \cdot 2 = 2 \quad \checkmark$



10. 设 $A = \begin{pmatrix} 1 & 2 & 3 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$, $B = \begin{pmatrix} 2 & 2 & 4 \\ 1 & 1 & 2 \\ 0 & 0 & 0 \end{pmatrix}$. 求可逆阵 P . 设 $P^{-1}AP = B$.

【解】: $A: 1, 0, 0$
 ξ_1, ξ_2, ξ_3
 P_1

$B: 1, 0, 0$
 η_1, η_2, η_3
 P_2

$(A - \lambda E) \cdot X = 0$

$\begin{cases} P_1^{-1}AP_1 = \Lambda = \begin{pmatrix} 1 & & \\ & 0 & \\ & & 0 \end{pmatrix} \\ P_2^{-1}BP_2 = \Lambda = \begin{pmatrix} 1 & & \\ & 0 & \\ & & 0 \end{pmatrix} \end{cases} \Rightarrow P_1^{-1}AP_1 = P_2^{-1}BP_2$
 $\Rightarrow \underline{P_2 P_1^{-1}} \cdot A \cdot \underline{P_1 P_2^{-1}} = B$

$P^{-1} = (\underline{P_1 P_2^{-1}})^{-1} = P_2 \cdot P_1^{-1}$. 取 $\underline{P} = \underline{P_2 \cdot P_1^{-1}}$ ✓

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